IMO Training 2018–19

Phase 2 (Level II)

Invariants and Monovariants

2018/10/13

Examples

1. There are n lamps, initially all off. Each lamp is equipped with a switch. When pressed, a lamp off will be turned on and a lamp on will be turned off. One is allowed to press exactly 100 switches each time.

   (a) When $n = 2018$, is it possible to have all the lamps on at the same time?

   (b) When $n = 2019$, is it possible to have all the lamps on at the same time?

2. On an island there are 13 red birds, 15 yellow birds and 17 green birds. When two birds of different colours meet, both of them change into the third colour. Is it possible for all birds to change into the same colour?

3. In the IMO training programme, each student has exactly three friends (friendship is mutual). Prove that it is possible to divide the students into two levels so that each student finds at most one friend in the same level.

4. (IMO 1986)

   To each vertex of a regular pentagon an integer is assigned in such a way that the sum of all the five numbers is positive. If three consecutive vertices are assigned the numbers x, y, z respectively and $y < 0$ then the following operation is allowed: the numbers x, y, z are replaced by $x+y$, $-y$ and $z+y$ respectively. Such an operation is performed repeatedly as long as at least one of the five numbers is negative. Determine whether this procedure necessarily comes to an end after a finite number of steps.

Exercise

5. There are 2018 cards, numbered 1 to 2018, placed in a row in order. In each step one swaps two adjacent cards. Is it possible that the cards are in their initial order after exactly 2019 steps?

6. (Modified from Putnam 2008)

   On the board there are finitely many positive integers written. Each time you are allowed to erase two integers provided that none of them is a multiple of the other, and then replace them by their L.C.M. and H.C.F. respectively. Is it possible for this process to continue forever?

7. Some (finitely many) stones are placed on the real number line, each at a lattice point. At each step, one is allowed to choose a point with at least two stones, and then move one stone from that point to the left by 1 unit and another from that point to the right by 1 unit. Is it possible to return all the stones to the initial position after a finite (positive) number of steps?

8. A positive integer starts with 101010… and each digit from the seventh is equal to the sum of the previous six digits modulo 10. (Thus the integer reads 101010350985…) Does the integer contain 010101 as consecutive digits somewhere (provided it has sufficiently many digits)?

9. (Modified from USAMO 1997)

   Find all positive real numbers x_0 for which there exists an infinite sequence x_0, x_1, x_2, … of real numbers satisfying the recurrence relation

   $$x_n = \left\{ \frac{p_n}{x_{n-1}} \right\} \quad \text{for all positive integers } n.$$

   (Here $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of *x*. and p_n denotes the *n*-th prime number.)

10. (Modified from IMO 2000)

    Let $n \geq 2$ be a fixed positive integer. Initially, *n* real numbers (not all the same) are written on the board. Each time, one is allowed to pick two real numbers *a* and *b* on the board with $a < b$, and then replace *a* by $b + \lambda(b - a)$, where λ is a fixed positive real number. Determine all values of λ for which you can guarantee (regardless of the choice of the initial numbers) all numbers on the board to exceed 10^{100} after finitely many steps.